HAROL MAXILAN

DATA ENGINEERING INTERN | DATA ANALYTICS

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Vavuniya, Sri Lanka

Professional Summary

Data Science undergraduate at Sabaragamuwa University of Sri Lanka with hands-on project experience in machine learning, natural language processing, computer vision, and AI-driven applications. Experienced in developing and evaluating classification models, experimenting with transformer architectures such as BERT and DistilBERT, handling class imbalance, and building data processing and ML pipelines using Python. Interested in developing reliable, practical AI solutions, improving model generalization, and applying machine learning to real-world problems.

Education

B.Sc. (Hons) in Data Science | 2024 – 2028 (Expected), *Sabaragamuwa University of Sri Lanka — Faculty of Computing*
B.Sc. (Hons) in Data Science | 2024–2028 (Expected)
Sabaragamuwa University of Sri Lanka – Faculty of Computing

Relevant Coursework: Database Systems, Data Warehousing, Data Science, Statistics, Computer Programming, Machine Learning, Artificial Intelligence

Awards

Nexora Ideathon : 3rd Place
Sabaragamuwa University of Sri Lanka

Cipher Hackathon : Finalist
IIT

J'PuraXtreme 3.0 Hackathon : Finalist
University of Sri Jayewardenepura

CodeRally 7.0 Hackathon : Finalist
IIT

Code Arena : Finalist
KDU

Projects

NLP-Based Emotion Detection Pipeline,
Technologies: Python, Scikit-learn, Hugging Face Transformers, BERT, DistilBERT

- Developed an end-to-end NLP classification pipeline combining a TF-IDF and Logistic Regression baseline with transformer-based models, including BERT and DistilBERT.
- Experimented with class imbalance handling techniques, including class weights, a custom WeightedTrainer, and label smoothing to improve minority-class recall.
- Evaluated models on an unseen test dataset using precision, recall, Macro F1, and Weighted F1 to compare performance and assess generalization.

Image-Based Emotion Detection, *Technologies: Python, OpenCV, Deep Learning, Streamlit, h5py*

- Developed a computer vision-based emotion recognition system using OpenCV for facial image processing.
- Trained and evaluated a deep learning model to recognize multiple emotional expressions from facial images, saving model weights in HDF5 format.
- Built an interactive Streamlit application to support image-based and real-time emotion detection workflows.

Friday — Automated Machine Learning Pipeline, *Technologies: Python, Pandas, Scikit-learn, FastAPI, Google Generative AI*

- Developed an automated machine learning pipeline with a FastAPI backend to support data-driven predictive modelling.
- Built data-processing workflows using Pandas and Scikit-learn to transform raw datasets into statistical analyses and meaningful insights.
- Integrated Google Generative AI to generate analytical reports from processed data.

TECHNICAL SKILLS

Programming Languages
Python, SQL, Java, JavaScript, PHP

Machine Learning & Deep Learning
Scikit-learn, PyTorch, model training, classification, regression, feature engineering, cross-validation, hyperparameter tuning, model evaluation

Natural Language Processing (NLP)
Hugging Face Transformers, BERT, DistilBERT, TF-IDF, text classification, text preprocessing, class imbalance handling

Computer Vision
OpenCV, image processing, facial emotion recognition, image classification, deep learning for image-based tasks

Data Science & Analysis
Pandas, NumPy, Matplotlib, exploratory data analysis (EDA), statistical analysis, data preprocessing

Databases
MySQL, MariaDB

Development & Tools
Git, GitHub, Jupyter Notebook, VS Code, Streamlit

AI Integration
Google Generative AI, AI-assisted application development

References

Ms.T.Kayalvizhy, *Lecturer,*
Sabaragamuwa University of Sri Lanka
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Mr. Mahesh Yogarajan, *CEO,*
Lanka.Tax, 247D, R. A. De Mel Mawatha, Colombo 3.
mahesh@yogarajan.com, +94-777-739-739

Leadership & Volunteer Experience

AWS Logistics and Operations Lead
10/2026 – Present
Sabaragamuwa University of Sri Lanka | Current

Club Cricket Team Captian
10/2025
VSC, Vavuniya

Clash of Pawns Chess Competition Mentor
06/2025
AUSDAV, Vavuniya

Zone C University Ambassador – IEEE INSL
04/2025
Sabaragamuwa University of Sri Lanka

Pentathalon 2.0 Logistics Lead
12/2023
AUSDAV, Vavuniya